

- Σ is a unitary map
- Each M_a is proportional to a single unitary matrix
- A quantum channel is inverted by another quantum channel only if it is unitary
- Decoherence is irreversible → Only when A and A' have same dimensions
- Decoherence can be reversed if and only if Knill-Laflamme Condition is satisfied

$\sum_i M_{ij}^\dagger M_{ij} = c_{ij} I \rightarrow$ Any unitary operation can be reversed

→ Max No. of Kraus Operators
 Σ acts on $H \rightarrow d$ dimensions

$$1 \leq N \leq d^2$$

$$\mathcal{E}(\rho) = \sum_{k=1}^N E_k \rho E_k^\dagger$$

